

Prof. Dr. Lincy Pyl and Selection Committee
Department of Mechanics of Materials and Constructions (MeMC)
Vrije Universiteit Brussel

Subject: Motivation letter – PhD candidate, Mechanics of Materials and Constructions (M²S)
Introduction video: <https://drive.google.com/file/d/14s-V07z3fopEllnUP8vmMJPMxQW57mg9/view>

Dear Prof. Pyl and Selection Committee,

I am Pooja Awasare, a civil and environmental engineer, and I am applying for the PhD position in Mechanics of Materials and Constructions at VUB. I want to continue into research on materials and structures in a group where laboratory testing and clear engineering questions sit at the centre of the work.

My Diploma and Bachelor's in Civil Engineering in India gave me a lot of time in the materials laboratory with concrete, steel, soil, aggregates and water. Those labs stayed with me for a simple reason: when a result looked wrong, I did not like to leave it at the number on the page. I wanted to understand what had caused it, and that habit of staying with a question until the reason becomes clear is still how I work. I am consistent in that way, and curious enough to keep reading, checking and comparing until things make sense.

I later completed a Master's in Environmental Engineering at HTW Dresden, where my thesis looked at how construction teams can manage projects and use resources more carefully. Through that work it became clearer to me that sustainable construction cannot skip materials and structures, because they are the heart of building. If we do not understand how materials behave under load and how structures use them, then much of the talk about sustainability stays thin.

During my time in Belgium I worked with a construction partner on circular practices in day-to-day project delivery, which showed me how industry thinks about waste, reuse and process on real projects. It also made clear what I still want to deepen: the mechanical side of materials and structures, through proper testing and careful analysis inside a research group.

I have been following MeMC and reading about your group's work, including Prof. Pyl's research on lightweight structures and on the mechanical performance of materials, where experiments are linked with analytical and numerical models. I have also followed MeMC's work on resource-efficient and circular building systems. What draws me to the group is the seriousness of those questions: how materials perform, how structures can use less material without losing strength, and how careful testing supports better design. That kind of depth matches the way I like to learn.

I know I am still early in a research career, but I am willing to prepare applications for external funding such as FWO and to learn what the group needs from a PhD student, including teaching support where asked. I would be glad to discuss a PhD topic with you on the mechanical behaviour of construction materials and more careful structural design for sustainability. Thank you for considering my application and for watching my introduction video.

Yours sincerely,

Pooja Awasare